UNIT IV

Design of Gear Boxes

speed reducers | multi-speed gear boxes for machine tools | structural diagrams | ray diagrams | automobile gearbox

I. Introduction

A **gear box** is a closed housing containing gears, shafts, bearings and lubricant that **changes speed and torque** between a prime mover and a machine. Two major types:

Type	Purpose	Examples
Speed reducer (fixed ratio)	Reduces motor speed to the machine speed with a constant ratio and increases torque	Conveyors, mixers, cranes, hoists, mills
Multi-speed gear box	Gives a **number of different speeds** (stepped) from a constant-speed motor	Machine tool headstocks (lathe, milling, drilling)
Automobile gear box	Selects torque multiplication for starting / hill / cruising; reverse & neutral	Sliding-mesh, constant-mesh, synchromesh, automatic

II. Design of speed reducers

Gear type	Ratio per stage	Efficiency	Remark
Spur / helical	4 - 6 (spur), 6 - 10 (helical)	97-99%	Parallel shafts; helical quieter, thrust
Bevel (straight / spiral)	up to 5 - 6	96-98%	Right-angle drive; often first stage
Worm	5 - 60 (single stage)	50-92%	Right angle, compact, silent; self-locking possible
Planetary (epicyclic)	3 - 10 per stage	97-98%	Coaxial, compact, high torque density

Types of reducers by number of stages

> **Single reduction**: one gear pair - $i \leq 6$-8. **Double reduction**: two pairs - i up to 30-40 (most common). **Triple**: i up to 100+.

> **Layouts**: in-line (coaxial) - input and output shafts on the same axis; **unfolded** (all shafts in one plane); **bevel-helical** (right angle, first bevel then helical); **worm-helical**.

> **Splitting of ratio**: for a double reduction, $i = i1 \times i2$ with **$i1 = 1.2$ to $1.4 \times i2$** (first, high-speed stage has smaller torque so can take the larger ratio; helps in oil-dip lubrication of both gears). For three stages take $i1 > i2 > i3$ (each about 1.2-1.3 times the next), so the low-speed gears are not oversized.

> Minimum centre distance / volume is found when each stage is designed for **equal strength and wear**, which gives $i1/i2 \approx 1.3$ for typical conditions.

Design procedure for a speed reducer

1. Collect: motor power P, speed N1, output speed N2 (overall $i = N1/N2$), duty, service factor, space.
2. Decide the number of stages & the **ratio split**; determine speeds & torques of each shaft with efficiencies (0.97-0.98 per stage including bearings):

$$T_{out} = 9.55 \times 10^6 \, P \, \eta / N_{out} \ \ (N{\cdot}mm); \quad \eta_{overall} = \eta_{gear}^{n} \times \eta_{bearing}^{m}$$

3. Design the **gear pairs** (Unit II/III): material, module, teeth, face width; check bending and wear for each stage.
4. Design **shafts** for torque + bending (gear forces Ft, Fr, Fa) using ASME code; select keys / splines.
5. Select **bearings** for each shaft from L10h life (e.g. 10,000-20,000 h industrial; 3,000-5,000 h automotive).
6. Design the **housing**, covers, lubrication, seals, breathers; check **heat dissipation**.
7. Prepare layout: gear/bearing positions, shaft steps, shoulders and circlips; assembly dimensions.

Example 1 - shaft speeds & torques of a double reducer

> Motor 7.5 kW at 1440 rpm; output about 90 rpm -> $i = 16$. Split: $i1 = 1.25\ i2$ -> $i2 = \sqrt{(16/1.25)} = 3.58$, $i1 = 4.47$. Efficiency 0.98 per stage.

> Shaft I: $N = 1440$ rpm, $T1 = 9.55 \times 10^6 \times 7.5/1440 = 49{,}740$ N·mm.

> Shaft II: $N = 1440/4.47 = 322$ rpm, $T2 = T1 \times i1 \times 0.98 =$ **217,900 N·mm**. Shaft III: $N = 322/3.58 = 90$ rpm, $T3 = T2 \times i2 \times 0.98 =$ **764,400 N·mm**.

> Output power = $T3\ N/9.55 \times 10^6 = 7.2$ kW (overall $\eta = 0.98^2 = 0.96$ ✓). Each stage gear pair is now designed from T1, T2 (Unit II/III method).

Example 1 (contd.) – first-stage gear and bearing on shaft 1

- Stage 1: z1 = 20, z2 = 89 (i1 = 4.45), 20° FD, C45, σb = 140 MPa, Cs = 1.25, ordinary cut gears. Design torque 49,740 × 1.25 = 62,175 N·mm; y = 0.1084; b = 10 m.
- m^3 Cv = 2T/[z1 σb (b/m) π y] = 124,350/[20 × 140 × 10 × π × 0.1084] = 13.0. m = 3: v = 4.52 m/s, Cv = 0.399, m^3Cv = 10.8 ✗ ; m = 4: v = 6.03, Cv = 0.332, m^3Cv = 21.2 ✓ -> **m = 4 mm**.
- d1 = 80, d2 = 356 mm, a = 218 mm, b = 40 mm. Ft = 2 × 49,740/80 = 1243 N ; Fd = Cs Ft/Cv = 4682 N ; Sb = 140 × 4 × 40 × 0.3405 = 7627 N ✓. Wear: Q = 1.63 -> K ≥ 4682/(40 × 1.63 × 80) = 0.90 -> **BHN ≥ 237** (take 240).
- Housing wall δ = 0.025 × 218 + 3 = 8.5 -> 9 mm. Bearing on shaft 1: Fr = Ft tan20° = 452 N, resultant 1323 N, central pinion -> 660 N per bearing; life 20,000 h at 1440 rpm: C = P (60 n L10h/10^6)^(1/3) = 0.66 × 12 = 7.9 kN -> with shock 1.3: 10.3 kN -> choose a 6205 deep-groove (C about 14 kN, check catalogue).

Shaft design (ASME code) in the gear box

- d^3 = (16/(π τ_allow)) √[$(K_b M)^2 + (K_t T)^2$] with τ_allow = 0.18 σut or 0.3 σy (lower; 25% less if keyway), Kb = 1.5–2 (suddenly applied, minor shock), Kt = 1–1.5.
- M = resultant bending moment from gear forces in two planes: M = √($M_h^2 + M_v^2$); axial thrust adds a moment Fa d/2 (helical, bevel, worm).
- Check deflection (< 0.0005 rad at gear) and critical speed; use standard sizes & fillet radius ≥ 0.1 d to cut stress concentration.
- Materials: C45 / 40C8 normalised or 40Cr1, 20MnCr5 for pinion shafts (integral with gear), ground fits h6/k6 at bearings.

Housing, lubrication & accessories

Item	Practice
Housing material	Grey cast iron FG200/FG260 (damping, cheap); welded steel for single piece; aluminium for light, small gearboxes.
Wall thickness	δ ≈ 0.025 a + 3 mm (a = centre distance, min 8 mm); ribs around bearing bosses; split along shaft axes for assembly.
Lubrication	**Splash / oil bath** for gear pitch velocity v < 12 m/s (oil level one tooth height for the lowest gear); **forced spray** for v > 12 m/s with pump, filter, cooler.

Item	Practice
Oil grade	ISO VG 150-320 (EP) for industrial reducers; SAE 80W-90 GL-4 for automobile gears; hypoid GL-5.
Accessories	**Breather** (pressure relief), oil level gauge, drain plug (magnetic), filler / inspection cover, lifting lugs, dowel pins, gaskets, oil seals.
Sealing	Lip seals, labyrinth, V-ring, O-ring on covers; bearing end caps with shims for end float.
Thermal	Heat generated Hg = P (1 - η); dissipated Hd = C A (t_oil - t_a) with C = 12-18 W/m²°C (natural), A = housing area; t_oil ≤ 80-90 °C; use fins/fan/oil cooler.

III. Multi-speed gear boxes for machine tools

A machine tool (lathe, drilling, milling) needs **different spindle speeds** for different cutting speed/ diameter/ tool material, from a constant-speed motor. The multi-speed gear box gives **z stepped speeds** N1 < N2 < ... < Nz between Nmin and Nmax.

Choice of progression of spindle speeds

Series	Law	Remark
Geometric progression (GP)	N2 = φ N1, N3 = φ² N1, ... Nk = N1 φ^(k-1)	**Most used**: constant % loss of cutting speed between steps; fewer gears; uses preferred numbers
Arithmetic progression (AP)	N_k = N1 + (k - 1) d	Too many closely spaced high speeds; wasteful of time (not used)
Harmonic progression (HP)	1/N in AP	Used in some special / feed boxes
Logarithmic	log N in AP	GP is the same as logarithmic progression

> **Speed range** Rn = Nmax/Nmin = φ^(z-1) -> **φ = (Nmax/Nmin)^(1/(z-1))** ; z = 1 + log Rn / log φ.

> **Percentage loss of cutting speed** (maximum, just above a step) = (φ - 1)/φ × 100. Eg φ = 1.26 ->

20.6%, 1.41 -> 29%, 1.12 -> 10.7%.

> Why GP? The economic loss in cutting speed is the **same percentage** at every speed step -> best compromise between number of steps and loss.

Standard ratio φ (preferred number series)	Series	Ratio	Max speed loss
R5	10^(1/5)	φ = 1.58	36.7%
R10	10^(1/10)	**φ = 1.26**	20.6%
R20	10^(1/20)	**φ = 1.12**	10.7%
R20/3	(R20 every 3rd)	**φ = 1.41**	29%
R40	10^(1/40)	φ = 1.06	5.7%
R10/2	(R10 every 2nd)	φ = 1.58	36.7%

Standard spindle speeds (R10, φ = 1.26) in rpm

> 31.5, 40, 50, 63, 80, 100, 125, 160, 200, 250, 315, 400, 500, 630, 800, 1000, 1250, 1600, 2000 … (IS 1076 preferred numbers; every 10th term is 10 × the previous decade).

> Eg. 16-speed lathe: Nmin = 31.5, Nmax = 1000 -> Rn = 31.7, φ = 31.7^(1/15) = 1.26 ✓ (R10).

IV. Kinematic layout of the gear box

> **Sliding gear clusters** (cones of 2 or 3 gears) on splined shafts give the stepped ratios; only **one pair** of a cluster is in mesh at a time. Also clutches and Norton tumblers (feed boxes).

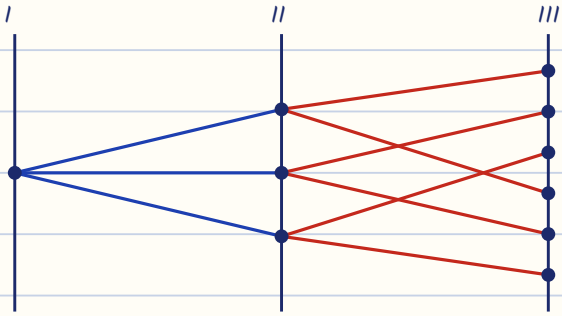
> Number of speeds z = p1 × p2 × p3 … where p = number of gear pairs (transmissions) in a group; p = 2, 3 (4 only with care). Eg 12 = 3 × 2 × 2.

> Total gear pairs = p1 + p2 + p3 …, shafts = number of groups + 1 ; gears = 2 × (p1 + p2 + …).

> Constant centre distance in each group: all pairs have the **same sum of teeth** Zs = z_driver + z_driven (same module).

Structural formula and structural diagram

Structural formula: z = p1 (x1) · p2 (x2) · p3 (x3) … where the number in brackets is the **characteristic x** of each group = **number of steps (in terms of φ) between adjacent ratios of that group**.



Structural formula

z = 3(1) × 2(3)

Shows only the **pattern** of ratios (no speed values), equal spacing.

Rule / quantity	Meaning
Normal ('basic') structure	x1 = 1 ; x2 = p1 ; x3 = p1 p2 ; ... (all speeds distinct, no overlap)
Range of a group	$R_g = \varphi^{(x(p-1))}$
Limits	Ratio of any gear pair: **1/4 ≤ i ≤ 2** (min 1/4: reduces size & noise; max 2 : avoids high speeds on spindle)
Maximum group range	$R_g \leq 8$ i.e. $\varphi^{(x(p-1))} \leq 8$ (ratio 4 : 1/2)
Ordering of groups	Characteristics generally increase toward the spindle; last group has the largest range -> intermediate shafts rotate fast & torque is small (smaller gears)
Choice of structure	Smallest weight/ size: reduce speeds **late** (speeds high on intermediate shafts, low torque)

z	Structural formulas (possible)	Group ranges ($\varphi^{x(p-1)}$)
4	2(1)·2(2)	φ ; φ^2
6	**3(1)·2(3)** ; 2(1)·3(2) ; 2(3)·3(1)	φ^2 ; φ^3
8	2(1)·2(2)·2(4)	φ ; φ^2 ; φ^4
9	3(1)·3(3)	φ^2 ; φ^6
12	**3(1)·2(3)·2(6)** ; 2(1)·3(2)·2(6) ; 2(1)·2(2)·3(4)	$\varphi^2,\varphi^3,\varphi^6$; $\varphi,\varphi^4,\varphi^6$; $\varphi,\varphi^2,\varphi^8$
16	2(1)·2(2)·2(4)·2(8)	φ ; φ^2 ; φ^4 ; φ^8

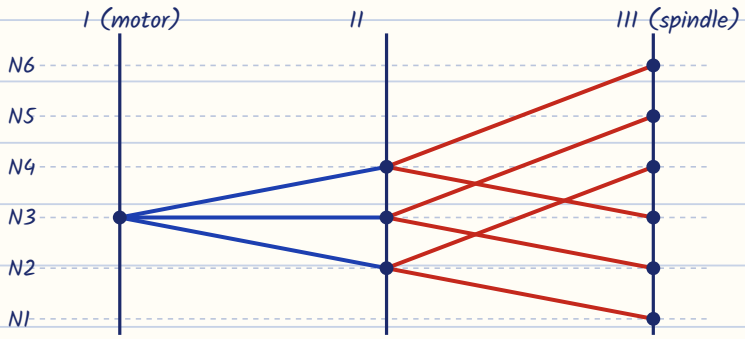
> Check: for φ = 1.26 (R10) 3(1)·2(3)·2(6) -> group 3 range $\varphi^6 = 4$ ✓ ≤ 8. For φ = 1.41, $\varphi^6 = 8$ ✓ borderline; $\varphi^8 = 16$ ✗ so 2(1)·2(2)·3(4) is not allowed for φ = 1.41.

> A **structural diagram** shows the pattern with equally spaced nodes (not to speed scale); a **ray**

(speed) diagram puts the shafts at actual speed levels (log scale).

V. Ray diagram

A **ray diagram** (speed diagram) is a graph with **vertical lines for shafts** and **horizontal lines for speeds on a logarithmic scale**, equally spaced by log φ. Each ray joins the speed of the driver to the speed of the driven shaft: **rising ray** = speed increase ($i > 1$), **falling ray** = reduction ($i < 1$), **horizontal** = $i = 1$. The number of steps up/down = log i / log φ.



Ray diagram (6 speeds, φ = 1.41)

Vertical lines = shafts;

horizontal lines = speeds on a log scale (equal spacing = φ).

Sloping up = speed ↑ ($i > 1$),

down = speed ↓ ($i < 1$).

Blue: first group (3 ratios, step φ^1).

Red: second group (2 ratios, step φ^3).

Rules for drawing ray diagrams

1. Draw as many vertical lines as shafts (motor, I, II, ..., spindle) and horizontal lines equal to the number of speeds, with spacing log φ.
2. Mark the spindle speeds N1 .. Nz on the last vertical line; mark the motor (or belt-reduced) speed on the first.
3. Start from the first group with the number of rays = p1, then the next group, so that all z spindle speeds appear (use the structural formula).
4. Ray limits: a rising ray may not exceed $i = 2$ (φ = 1.41: 2 steps up; φ = 1.26: 3 steps up); a falling ray may not go below $i = 1/4$ (φ = 1.41: 4 steps down; φ = 1.26: 6 steps down).
5. Keep intermediate-shaft speeds as **high as possible**; and the spindle's last transmission can be the largest reduction.

Example 2 – 6-speed gear box (spindle 100 – 560 rpm, motor via belt to 200 rpm)

> z = 6, Nmin = 100, Nmax = 560 -> Rn = 5.6, φ = 5.6^(1/5) = 1.41 (R20/3). Speeds: **100, 140, 200, 280, 400, 560 rpm**. Structural formula **3(1)·2(3)**.

> Group 1 (3 ratios, x = 1): ratios 1/φ, 1, φ -> shaft II speeds 140, 200, 280. Group 2 (2 ratios, x = 3): ratios 1/φ and φ² (= 2) -> from each shaft-II speed to a lower (÷1.41) and a higher (× 2) spindle speed.

> Check limits: ratios 0.71, 1, 1.41, 2 all lie between 1/4 and 2 ✓ ; group 2 range = φ³ = 2.8 ≤ 8 ✓.

Determination of number of teeth

1. For a group with sum 2s, a ratio i = a/b (as a ratio of coprime integers) gives driver z = 2s a/(a + b) ; driven 2s - z.
2. Choose 2s such that all driver/driven teeth are integers ≥ **17–20** (minimum on the pinion), and ≤ 100–120; ratios must match φ-series within ± 10 (φ - 1) %.
3. Check actual output speeds, tooth strength (design the smallest pinion of each group) and module (same m in the group).

Teeth for Example 2

> Group 1: 2s = 72: i = 0.707 -> 30/42 (0.714); i = 1 -> 36/36; i = 1.41 -> 42/30 (1.40). Group 2: 2s = 84: i = 0.707 -> 35/49 (0.714); i = 2 -> 56/28 (2.0).

> Speeds: shaft II = 200 × (0.714, 1, 1.4) = 143, 200, 280; spindle = {143 × 0.714, 143 × 2, 200 × 0.714, 200 × 2, 280 × 0.714, 280 × 2} = **102, 143, 200, 286, 400, 560 rpm** (errors ≤ 2% < 4.1% ✓).

> Gears total = 2 (3 + 2) = 10; shafts = 3 (I, II, spindle).

Example 3 – 12-speed gear box (R10, φ = 1.26)

Spindle speeds 40 to 500 rpm: **40, 50, 63, 80, 100, 125, 160, 200, 250, 315, 400, 500** (Rn = 12.5; φ = 12.5^(1/11) = 1.26). Structural formula **3(1)·2(3)·2(6)**. Input shaft I = 200 rpm (motor 1440 rpm through a belt drive).

Group	p (x)	Ratios (steps)	Numeric ratios	2s	Teeth driver / driven
I (I -> II)	3 (1)	φ^{-3}, φ^{-2}, φ^{-1}	0.50, 0.63, 0.79	72	24/48, 28/44, 32/40
2 (II -> III)	2 (3)	φ^{-1}, φ^{2}	0.79, 1.59	80	35/45, 49/31

Group	p (x)	Ratios (steps)	Numeric ratios	2s	Teeth driver / driven
3 (III -> spindle)	2 (6)	φ^{-3}, φ^{3}	0.50, 2.00	90	30/60, 60/30

> Speeds: shaft II = 100, 127, 160 ; shaft III = 78, 99, 124, 158, 201, 253 ; spindle = 39, 49.5, 62, 79, 100.5, 126, 156, 198, 249, 316, 402, 506 rpm - errors ≤ 2.8% (refine teeth to match ± 2.6% if required).

> Number of gears = 2 (3 + 2 + 2) = 14; shafts = 4. Group ranges: φ^{2} = 1.6, φ^{3} = 2, φ^{6} = 4 (all ≤ 8 ✓). All ratios between 0.5 and 2 ✓ ⊂ [1/4, 2].

> A 9-speed box is 3(1)·3(3): ranges φ^{2} and φ^{6} -> with φ = 1.41 the last group range is φ^{6} = 8 (limit).

Layout & design notes for machine-tool gear boxes

> **Sliding cluster**: gears in one cluster must have a difference of at least 4 teeth between adjacent gears to avoid fouling; axial width kept small; shifting fork at the groove.

> Use **two-gear clusters** (2 pairs) where possible: shorter shaft & housing; 3-gear cluster needs interlocking so two clusters cannot be engaged at once.

> Module is normally the **same for a group** (the same 2s), but can differ between groups: higher torque near the spindle -> larger module.

> **Shaft diameters** are found from the torque at the lowest speed with ASME formula; intermediate shaft diameter < spindle diameter; spindle of lathe has bore for bar and is stiff (deflection limit).

> **Bearings**: spindle has precision tapered roller or angular contact pair (preloaded); others ball bearings; lubrication by oil bath/ splash, with filter & sight glass.

> Efficiency: each gear pair about 0.97-0.98, each bearing pair 0.99; overall about 0.9 for a 3-group box.

Planetary (epicyclic) speed reducer

> Elements: sun gear, planets on a carrier, ring (annulus). Ratio (ring fixed, sun input, carrier output) = 1 + z_ring/z_sun = 3 to 10; coaxial input/output, several planets share the load (3-4) so power density is high.

> **Tabular method**: with the carrier fixed, rotate gears; then add the carrier's rotation. Used in automatic gearboxes, wheel hubs, winches.

> Assembly condition: (z_sun + z_ring)/n_planets = integer; z_ring = z_sun + 2 z_planet.

Gear box faults & maintenance

Symptom	Likely cause	Remedy
Noise / whine	Wrong backlash, worn or misaligned gears, low oil	Re-shim, replace, correct oil level
Overheating	Overload, too much or too little oil, wrong viscosity, blocked breather	Check load, oil grade/level, clean breather, add cooler
Oil leakage	Worn seals, loose cover bolts, clogged breather pressure	Replace seal/gasket, clean breather
Hard gear shifting	Worn synchroniser, clutch drag, wrong oil	Replace synchro, adjust clutch
Bearing failure	Contaminated oil, wrong preload/fit	Filter oil, correct fits, replace bearings

VI. Design of a machine-tool gear box – summary of steps

1. Fix Nmin, Nmax, z (from the work range), choose φ from the standard series; list spindle speeds.
2. Choose the **structural formula**; check group ranges (≤ 8) and ratios ($1/4 \le i \le 2$).
3. Draw **ray diagram**; read all gear ratios.
4. Select sums of teeth Zs and find tooth numbers; compute actual speeds and errors.
5. Compute power/torque on each shaft (lowest speed of a shaft governs the maximum torque), design the **weakest pinion** of each group for bending & wear (Units II-III), face width b = 10 m.
6. Design shafts (torque at the lowest speed), splines, keys, bearings, clutch/ cluster shifting, lubrication, housing.

> Power at the spindle is kept constant (constant-power drive) between steps: torque = P/ω is greatest at the **lowest spindle speed**.

> Gear-shifting: sliding cluster on splines with fork; interlock; **dog clutch / friction clutch** for fast change; in modern machines VFD (inverter) motors reduce steps.

VII. Automobile gear box (extra – relevant to the subject)

Type	Working	Features
Sliding mesh	Gears on splined shaft slide to engage countershaft gears	Simple, noisy, needs double declutching; spur / helical; old trucks
Constant mesh	All gears always in mesh; dog clutches select the gear	Helical gears, quieter, easier shift

Type	Working	Features
Synchromesh	Cone synchronisers equalise speeds before dog engagement	Smooth shift without double clutching; modern cars
Epicyclic / automatic	Planetary gear sets with clutches and bands (or CVT / DCT)	Torque converter, auto shifting

Choice of gear ratios in a vehicle gearbox

> **Top gear ratio** (overall) from maximum speed: N_engine at max power -> vehicle speed V = 0.377 N r_w/(G_total) (km/h, r_w in m).

> **First gear ratio** from **gradeability** / starting: engine torque × G1 × G_final × η / r_w ≥ W (sin α + f cos α); usually G1 = 3.5 - 4.5, reverse about 3.5 - 4.

> **Intermediate gears in GP**: G_k = G1 (G_top/G1)^((k-1)/(n-1)) - gives equal % speed change on shifts. Eg 4-speed G1 = 4, top = 1: G = 4, 2.52, 1.59, 1.0.

> Practice: ratios are slightly 'closer' (progressive) at high gears for better acceleration; 5th gear is an overdrive (0.8).

> Gearbox rated at maximum engine torque × G1 ; helical gears 20° normal, β = 20°-30°; gear material 20MnCrS case-hardened, shaved; shafts splined; bearing life about 3000-5000 h.

VIII. Quick revision & important questions

Remember

> Speed reducer: i1 ≈ 1.3 i2 for double reduction; T_out = 9.55 × 10^6 P η/N.

> GP: N_k = N1 φ^(k-1); φ = (Nmax/Nmin)^(1/(z-1)); standard φ: 1.12, 1.26, 1.41, 1.58, 2.

> Structural formula z = p1(x1)·p2(x2) ...; x1 = 1, x2 = p1, x3 = p1 p2; group range φ^(x(p-1)) ≤ 8; ratio 1/4 ≤ i ≤ 2.

> Ray diagram: shafts vertical, speeds horizontal (log scale). Teeth from sum Zs and ratio a/b.

1. Explain the design procedure of a two-stage speed reducer. (10 M)
2. Why geometric progression is used for spindle speeds? Derive φ. (5 M)
3. Explain structural formula and the rules for ray diagrams. (10 M)
4. Design a 9-speed (or 12-speed) gear box for a lathe: structural & ray diagrams, teeth of gears. (10 M)

5. Determine the number of teeth on each gear of a 6-speed gear box. (10 M)
6. Explain the types of automobile gear boxes with neat sketches. (5 M)